18/07/2026

1. Description

1.1. Project

Project Name	Round_dialPin_config
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	07/18/2026

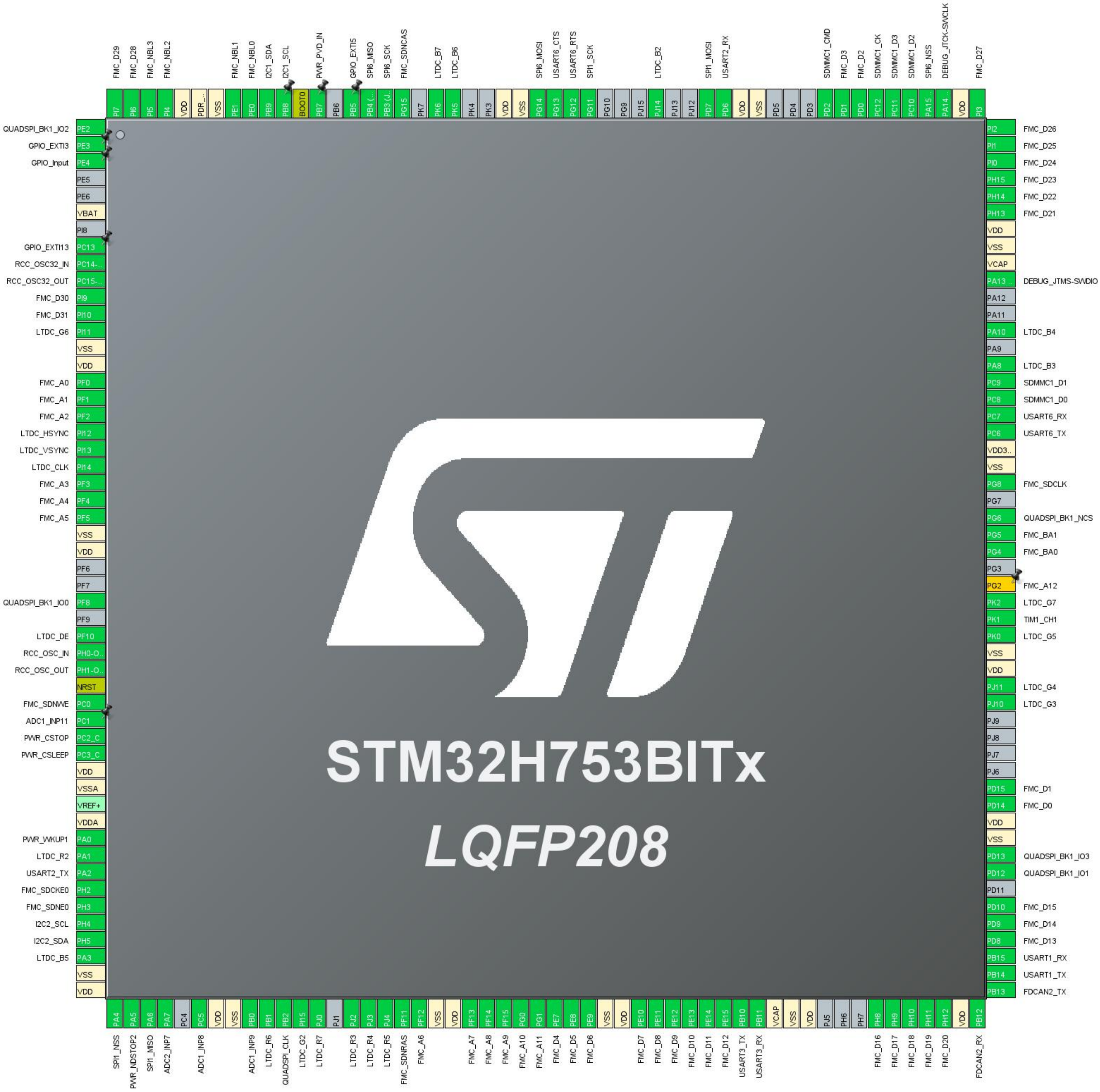
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H743/753
MCU name	STM32H753BITx
MCU Package	LQFP208
MCU Pin number	208

1.3. Core(s) information

Core(s)	ARM Cortex-M7

2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	QUADSPI_BK1_IO2	
2	PE3	I/O	GPIO_EXTI3	
3	PE4 *	I/O	GPIO_Input	
6	VBAT	Power		
8	PC13	I/O	GPIO_EXTI13	
9	PC14-OSC32_IN (OSC32_IN)	I/O	RCC_OSC32_IN	
10	PC15-OSC32_OUT (OSC32_OUT)	I/O	RCC_OSC32_OUT	
11	PI9	I/O	FMC_D30	
12	PI10	I/O	FMC_D31	
13	PI11	I/O	LTDC_G6	
14	VSS	Power		
15	VDD	Power		
16	PF0	I/O	FMC_A0	
17	PF1	I/O	FMC_A1	
18	PF2	I/O	FMC_A2	
19	PI12	I/O	LTDC_HSYNC	
20	PI13	I/O	LTDC_VSYNC	
21	PI14	I/O	LTDC_CLK	
22	PF3	I/O	FMC_A3	
23	PF4	I/O	FMC_A4	
24	PF5	I/O	FMC_A5	
25	VSS	Power		
26	VDD	Power		
29	PF8	I/O	QUADSPI_BK1_IO0	
31	PF10	I/O	LTDC_DE	
32	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
33	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
34	NRST	Reset		
35	PC0	I/O	FMC_SDNWE	
36	PC1	I/O	ADC1_INP11	
37	PC2_C	I/O	PWR_CSTOP	
38	PC3_C	I/O	PWR_CSLEEP	
39	VDD	Power		
40	VSSA	Power		

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
42	VDDA	Power		
43	PA0	I/O	PWR_WKUP1	
44	PA1	I/O	LTDC_R2	
45	PA2	I/O	USART2_TX	
46	PH2	I/O	FMC_SDCKE0	
47	PH3	I/O	FMC_SDNE0	
48	PH4	I/O	I2C2_SCL	
49	PH5	I/O	I2C2_SDA	
50	PA3	I/O	LTDC_B5	
51	VSS	Power		
52	VDD	Power		
53	PA4	I/O	SPI1_NSS	
54	PA5	I/O	PWR_NDSTOP2	
55	PA6	I/O	SPI1_MISO	
56	PA7	I/O	ADC2_INP7	
58	PC5	I/O	ADC1_INP8	
59	VDD	Power		
60	VSS	Power		
61	PB0	I/O	ADC1_INP9	
62	PB1	I/O	LTDC_R6	
63	PB2	I/O	QUADSPI_CLK	
64	PI15	I/O	LTDC_G2	
65	PJ0	I/O	LTDC_R7	
67	PJ2	I/O	LTDC_R3	
68	PJ3	I/O	LTDC_R4	
69	PJ4	I/O	LTDC_R5	
70	PF11	I/O	FMC_SDNRAS	
71	PF12	I/O	FMC_A6	
72	VSS	Power		
73	VDD	Power		
74	PF13	I/O	FMC_A7	
75	PF14	I/O	FMC_A8	
76	PF15	I/O	FMC_A9	
77	PG0	I/O	FMC_A10	
78	PG1	I/O	FMC_A11	
79	PE7	I/O	FMC_D4	
80	PE8	I/O	FMC_D5	
81	PE9	I/O	FMC_D6	
82	VSS	Power		

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
83	VDD	Power		
84	PE10	I/O	FMC_D7	
85	PE11	I/O	FMC_D8	
86	PE12	I/O	FMC_D9	
87	PE13	I/O	FMC_D10	
88	PE14	I/O	FMC_D11	
89	PE15	I/O	FMC_D12	
90	PB10	I/O	USART3_TX	
91	PB11	I/O	USART3_RX	
92	VCAP	Power		
93	VSS	Power		
94	VDD	Power		
98	PH8	I/O	FMC_D16	
99	PH9	I/O	FMC_D17	
100	PH10	I/O	FMC_D18	
101	PH11	I/O	FMC_D19	
102	PH12	I/O	FMC_D20	
103	VDD	Power		
104	PB12	I/O	FDCAN2_RX	
105	PB13	I/O	FDCAN2_TX	
106	PB14	I/O	USART1_TX	
107	PB15	I/O	USART1_RX	
108	PD8	I/O	FMC_D13	
109	PD9	I/O	FMC_D14	
110	PD10	I/O	FMC_D15	
112	PD12	I/O	QUADSPI_BK1_IO1	
113	PD13	I/O	QUADSPI_BK1_IO3	
114	VSS	Power		
115	VDD	Power		
116	PD14	I/O	FMC_D0	
117	PD15	I/O	FMC_D1	
122	PJ10	I/O	LTDC_G3	
123	PJ11	I/O	LTDC_G4	
124	VDD	Power		
125	VSS	Power		
126	PK0	I/O	LTDC_G5	
127	PK1	I/O	TIM1_CH1	
128	PK2	I/O	LTDC_G7	
129	PG2 **	I/O	FMC_A12	

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
131	PG4	I/O	FMC_BA0	
132	PG5	I/O	FMC_BA1	
133	PG6	I/O	QUADSPI_BK1_NCS	
135	PG8	I/O	FMC_SDCLK	
136	VSS	Power		
137	VDD33_USB	Power		
138	PC6	I/O	USART6_TX	
139	PC7	I/O	USART6_RX	
140	PC8	I/O	SDMMC1_D0	
141	PC9	I/O	SDMMC1_D1	
142	PA8	I/O	LTDC_B3	
144	PA10	I/O	LTDC_B4	
147	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
148	VCAP	Power		
149	VSS	Power		
150	VDD	Power		
151	PH13	I/O	FMC_D21	
152	PH14	I/O	FMC_D22	
153	PH15	I/O	FMC_D23	
154	PI0	I/O	FMC_D24	
155	PI1	I/O	FMC_D25	
156	PI2	I/O	FMC_D26	
157	PI3	I/O	FMC_D27	
158	VDD	Power		
159	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
160	PA15 (JTDI)	I/O	SPI6_NSS	
161	PC10	I/O	SDMMC1_D2	
162	PC11	I/O	SDMMC1_D3	
163	PC12	I/O	SDMMC1_CK	
164	PD0	I/O	FMC_D2	
165	PD1	I/O	FMC_D3	
166	PD2	I/O	SDMMC1_CMD	
170	VSS	Power		
171	VDD	Power		
172	PD6	I/O	USART2_RX	
173	PD7	I/O	SPI1_MOSI	
176	PJ14	I/O	LTDC_B2	
180	PG11	I/O	SPI1_SCK	
181	PG12	I/O	USART6_RTS	

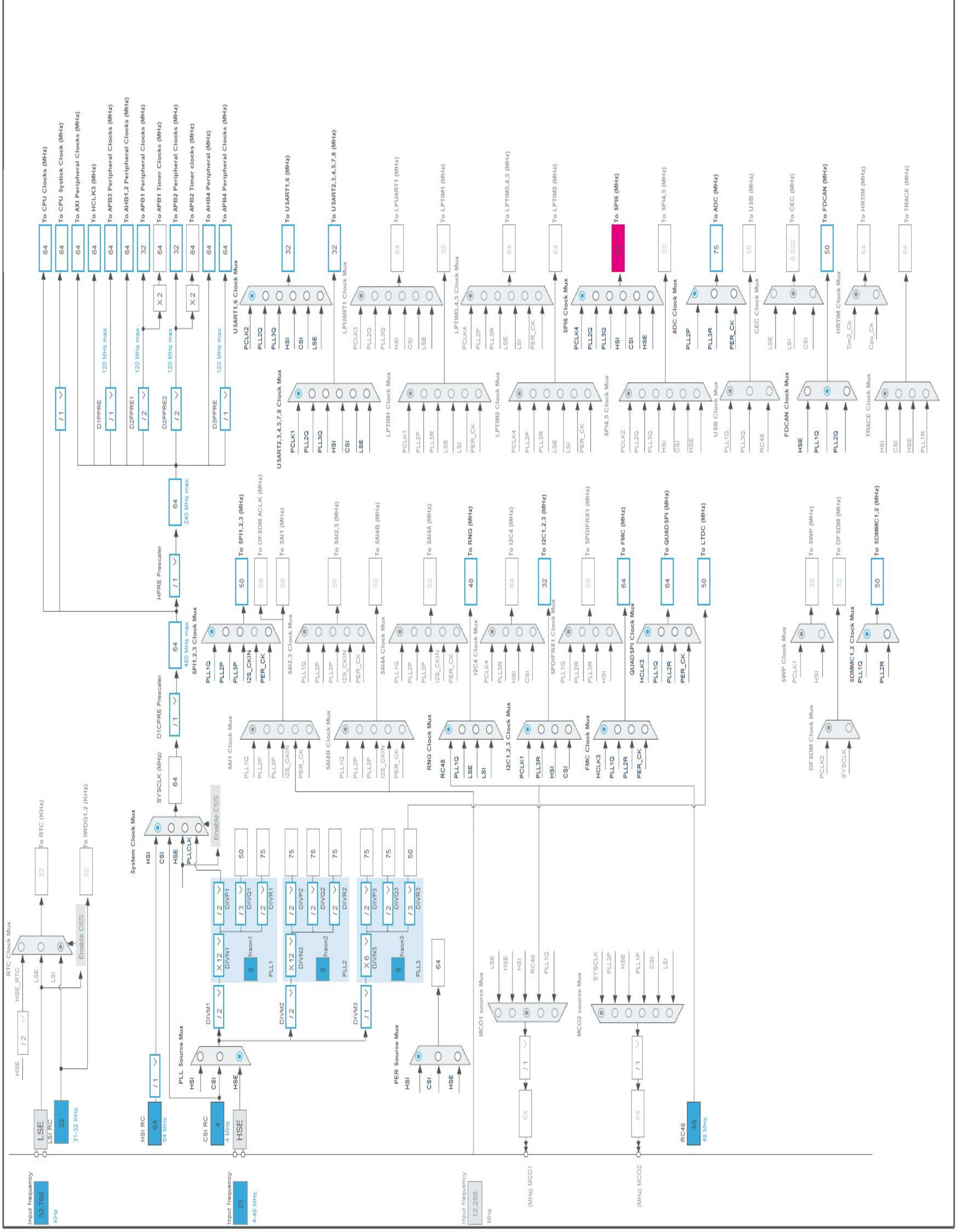
Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
182	PG13	I/O	USART6_CTS	
183	PG14	I/O	SPI6_MOSI	
184	VSS	Power		
185	VDD	Power		
188	PK5	I/O	LTDC_B6	
189	PK6	I/O	LTDC_B7	
191	PG15	I/O	FMC_SDNCAS	
192	PB3 (JTDO/TRACESWO)	I/O	SPI6_SCK	
193	PB4 (NJTRST)	I/O	SPI6_MISO	
194	PB5	I/O	GPIO_EXTI5	
196	PB7	I/O	PWR_PVD_IN	
197	BOOT0	Boot		
198	PB8	I/O	I2C1_SCL	
199	PB9	I/O	I2C1_SDA	
200	PE0	I/O	FMC_NBL0	
201	PE1	I/O	FMC_NBL1	
202	VSS	Power		
203	PDR_ON	Power		
204	VDD	Power		
205	PI4	I/O	FMC_NBL2	
206	PI5	I/O	FMC_NBL3	
207	PI6	I/O	FMC_D28	
208	PI7	I/O	FMC_D29	

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32H753BITx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H743/753
MCU	STM32H753BITx
Datasheet	DS12117_Rev7

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

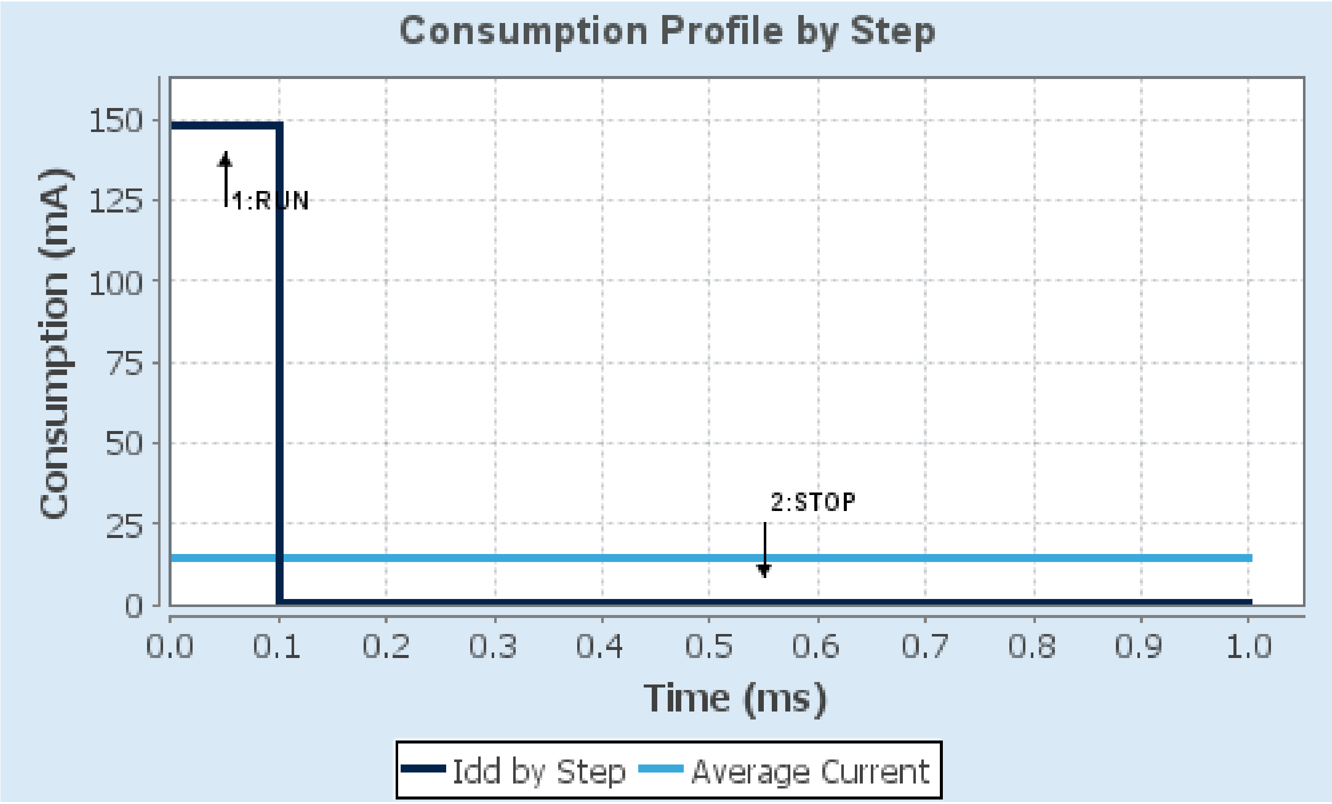
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 µA
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.91	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
150
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	Round_dialPin_config
Project Folder	D:\New folder\Round_dialPin_config\Documents\Round_dialPin_config
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_H7 V1.13.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	Yes

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_CRYP_Init	CRYP
4	MX_FMC_Init	FMC
5	MX_HASH_Init	HASH
6	MX_I2C1_Init	I2C1
7	MX_LTDC_Init	LTDC
8	MX_QUADSPI_Init	QUADSPI
9	MX_RNG_Init	RNG
10	MX_SDMMC1_SD_Init	SDMMC1
11	MX_SPI1_Init	SPI1

Rank	Function Name	Peripheral Instance Name
12	MX_TIM1_Init	TIM1
13	MX_USART1_UART_Init	USART1
14	MX_USART2_UART_Init	USART2
15	MX_USART3_UART_Init	USART3
16	MX_FATFS_Init	FATFS
17	MX_MBEDTLS_Init	MBEDTLS
18	MX_ADC1_Init	ADC1
19	MX_ADC2_Init	ADC2
20	MX_FDCAN2_Init	FDCAN2
21	MX_USART6_UART_Init	USART6
22	MX_CRC_Init	CRC
23	MX_I2C2_Init	I2C2
24	MX_SPI6_Init	SPI6

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN8

mode: IN9

mode: IN11

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 2
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Oversampling Ratio	1
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	Channel 8
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.2. ADC2
mode: IN7
3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 2
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Oversampling Ratio	1
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	Channel 7
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.3. CORTEX_M7

3.3.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode	**Enabled ***

Cortex Interface Settings:

CPU ICache	Disabled
CPU DCache	Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	**0x0 ***
MPU Region Size	4GB
MPU SubRegion Disable	**0x87 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	ENABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

3.4. CRC
mode: Activated
3.4.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.5. CRYP
mode: Activated
3.5.1. Parameter Settings:

Algorithm:

Data encryption algorithm	TDES ECB

Parameters:

Data type	32b(no swapping)

Encryption/Decryption key	00000000 00000000 00000000 00000000 00000000 00000000

3.6. DEBUG
Debug: Serial Wire

3.7. FDCAN2
mode: Activated
3.7.1. Parameter Settings:

Basic Parameters:

Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	0
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	0
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0
Tx Fifo Queue Elmts Nbr	0
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable

Bit Timings Parameters:

Nominal Prescaler	16
Nominal Time Quantum	**320.0 ***

Nominal Time Seg1	1
Nominal Time Seg2	1
Nominal Time for one Bit	**960 ***
Nominal Baud Rate	**1041666 ***

3.8. FMC

SDRAM 1

Clock and chip enable: SDCKE0+SDNE0

Internal bank number: 4 banks

Address: 12 bits

Data: 32 bits

Byte enable: 32-bit byte enable

3.8.1. SDRAM 1:

SDRAM control:

Bank	SDRAM bank 1
Number of column address bits	8 bits
Number of row address bits	12 bits
CAS latency	1 memory clock cycle
Write protection	Disabled
SDRAM common clock	Disabled
SDRAM common burst read	Disabled
SDRAM common read pipe delay	0 HCLK clock cycle

SDRAM timing in memory clock cycles:

Load mode register to active delay	16
Exit self-refresh delay	16
Self-refresh time	16
SDRAM common row cycle delay	16
Write recovery time	16
SDRAM common row precharge delay	16
Row to column delay	16

3.8.2. Bank Mapping:

Mapping parameters:

FMC bank mapping	Default mapping

3.9. HASH
mode: Activated
3.9.1. Parameter Settings:

Algorithm:

Secure hash algorithm type	SHA1

Parameters:

Hash data type in bit	32

3.10. I2C1
I2C: I2C
3.10.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x00707CBB ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.11. I2C2
I2C: I2C
3.11.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x00707CBB ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.12. LTDC

Display Type: RGB666 (18 bits)

3.12.1. Parameter Settings:

Synchronization for Width:

Horizontal Synchronization Width	8
Horizontal Back Porch	7
Active Width	640
Horizontal Front Porch	6
HSync Width	7
Accumulated Horizontal Back Porch Width	14
Accumulated Active Width	654
Total Width	660

Synchronization for Height:

Vertical Synchronization Height	4
Vertical Back Porch	2
Active Height	480
Vertical Front Porch	2
VSync Height	3
Accumulated Vertical Back Porch Height	5
Accumulated Active Height	485
Total Height	487

Signal Polarity:

Horizontal Synchronization Polarity	Active Low
Vertical Synchronization Polarity	Active Low
Data Enable Polarity	Active Low

Pixel Clock Polarity	Normal Input

Layer Default Color:

Red	0
Green	0
Blue	0

3.12.2. Layer Settings:

Layer Default Color:

Layer 0 - Alpha	0
Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	0
Layer 1 - Alpha	0
Layer 1 - Blue	0
Layer 1 - Green	0
Layer 1 - Red	0

Number of Layers:

Number of Layers	2 layers

Windows Position:

Layer 0 - Window Horizontal Start	0
Layer 0 - Window Horizontal Stop	0
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	0
Layer 1 - Window Horizontal Start	0
Layer 1 - Window Horizontal Stop	0
Layer 1 - Window Vertical Start	0
Layer 1 - Window Vertical Stop	0

Pixel Parameters:

Layer 0 - Pixel Format	ARGB8888
Layer 1 - Pixel Format	ARGB8888

Blending:

Layer 0 - Alpha constant for blending	0
Layer 0 - Blending Factor1	Alpha constant
Layer 0 - Blending Factor2	Alpha constant
Layer 1 - Alpha constant for blending	0
Layer 1 - Blending Factor1	Alpha constant
Layer 1 - Blending Factor2	Alpha constant

Frame Buffer:

Layer 0 - Color Frame Buffer Start Adress	0
Layer 0 - Color Frame Buffer Line Length (Image	0

Width)	
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	0
Layer 1 - Color Frame Buffer Start Adress	0
Layer 1 - Color Frame Buffer Line Length (Image Width)	0
Layer 1 - Color Frame Buffer Number of Lines (Image Height)	0

3.13. MEMORYMAP
mode: Activated

3.14. PWR
mode: Wake-Up 1
mode: Monitoring State
Power Voltage Detector In: Power Voltage Detector In (External input analog voltage)
3.14.1. Parameter Settings:

Programmable_Voltage_Detector_Settings:

PWR PVD Mode	basic mode is used

3.15. QUADSPI
QuadSPI Mode: Bank1 with Quad SPI Lines
3.15.1. Parameter Settings:

General Parameters:

Clock Prescaler	255
Fifo Threshold	1
Sample Shifting	No Sample Shifting
Flash Size	1
Chip Select High Time	1 Cycle
Clock Mode	Low
Flash ID	Flash ID 1
Dual Flash	Disabled

3.16. RCC
High Speed Clock (HSE): Crystal/Ceramic Resonator
Low Speed Clock (LSE) : Crystal/Ceramic Resonator
3.16.1. Parameter Settings:

Power Parameters:	
SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 3
RCC Parameters:	
TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64
System Parameters:	
VDD voltage (V)	3.3
Flash Latency(WS)	1 WS (2 CPU cycle)
Product revision	rev.V
PLL range Parameters:	
PLL1 clock Input range	Between 8 and 16 MHz
PLL2 input frequency range	Between 8 and 16 MHz
PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	MEDIUM VCO range
PLL2 clock Output range	MEDIUM VCO range
PLL3 clock Output range	MEDIUM VCO range

3.17. RNG
mode: Activated
3.17.1. Parameter Settings:

Clock Error Detection	Enable

3.18. SDMMC1
Mode: SD 4 bits Wide bus
3.18.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.19. SPI1
Mode: Full-Duplex Master
Hardware NSS Signal: Hardware NSS Input Signal
3.19.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	**25.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Input Hardware
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.20. SPI6
Mode: Full-Duplex Master
Hardware NSS Signal: Hardware NSS Input Signal

3.20.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	**32.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Input Hardware
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.21. SYS
Timebase Source: SysTick

3.22. TIM1
Channel1: Output Compare CH1
3.22.1. Parameter Settings:

Counter Settings:	
Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable

Output Compare Channel 1:

Mode	Frozen (used for Timing base)
Pulse (16 bits value)	0
Output compare preload	Disable
CH Polarity	High
CH Idle State	Reset

3.23. USART1

Mode: Asynchronous

3.23.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.24. USART2

Mode: Asynchronous

3.24.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration

Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.25. USART3

Mode: Asynchronous

3.25.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.26. USART6
Mode: Asynchronous
Hardware Flow Control (RS232): CTS/RTS
3.26.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.27. FATFS
mode: SD Card
3.27.1. Set Defines:

Version:

FATFS version	R0.12c

Function Parameters:

FS_READONLY (Read-only mode)	Disabled
FS_MINIMIZE (Minimization level)	Disabled
USE_STRFUNC (String functions)	Enabled with LF -> CRLF conversion

USE_FIND (Find functions)	Disabled
USE_MKFS (Make filesystem function)	Enabled
USE_FASTSEEK (Fast seek function)	Enabled
USE_EXPAND (Use f_expand function)	Disabled
USE_CHMOD (Change attributes function)	Disabled
USE_LABEL (Volume label functions)	Disabled
USE_FORWARD (Forward function)	Disabled

Locale and Namespace Parameters:

CODE_PAGE (Code page on target)	Latin 1
USE_LFN (Use Long Filename)	Disabled
MAX_LFN (Max Long Filename)	255
LFN_UNICODE (Enable Unicode)	ANSI/OEM
STRF_ENCODE (Character encoding)	UTF-8
FS_RPATH (Relative Path)	Disabled

Physical Drive Parameters:

VOLUMES (Logical drives)	1
MAX_SS (Maximum Sector Size)	512
MIN_SS (Minimum Sector Size)	512
MULTI_PARTITION (Volume partitions feature)	Disabled
USE_TRIM (Erase feature)	Disabled
FS_NOFSINFO (Force full FAT scan)	0

System Parameters:

FS_TINY (Tiny mode)	Disabled
FS_EXFAT (Support of exFAT file system)	Disabled
FS_NORTC (Timestamp feature)	Dynamic timestamp
FS_REENTRANT (Re-Entrancy)	Enabled
FS_TIMEOUT (Timeout ticks)	1000
USE_MUTEX	Disabled
SYNC_t (O/S sync object)	osSemaphoreId_t
FS_LOCK (Number of files opened simultaneously)	2

3.27.2. Advanced Settings:

SDIO/SDMMC:

SDMMC instance	SDMMC1
Use dma template	Enabled
BSP code for SD	Generic

3.28. FREERTOS

Interface: CMSIS_V2

3.28.1. Config parameters:

API:	
FreeRTOS API	CMSIS v2
Versions:	
FreeRTOS version	10.6.2
CMSIS-RTOS version	2.1.3
MPU/FPU:	
ENABLE_MPU	Disabled
ENABLE_FPU	Disabled
Kernel settings:	
USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
Memory management settings:	
Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	15360
HEAP_CLEAR_MEMORY_ON_FREE	0
Memory Management scheme	heap_4
Hook function related definitions:	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled

USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.28.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled

xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.28.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings (see parameter description first):

Use FW pack heap file	Enabled

3.29. MBEDTLS
mode: Enabled
3.29.1. Modes:

TCP/IP stack:

TCP/IP stack	None

RNG dependency:

RNG IP	HW RNG(ST entropy)

Modes:

MBEDTLS_SSL_CLI_C	Not Defined
MBEDTLS_SSL_SRV_C	Not Defined

3.29.2. Feature support:

System support:

MBEDTLS_HAVE_ASM	Defined
MBEDTLS_NO_UDBL_DIVISION	Defined
MBEDTLS_HAVE_TIME	Defined
MBEDTLS_HAVE_TIME_DATE	Defined

General:

MBEDTLS_ECP_NIST_OPTIM	Defined
MBEDTLS_ECDSA_DETERMINISTIC	Defined
MBEDTLS_PK_PARSE_EC_EXTENDED	Defined
MBEDTLS_ERROR_STRERROR_DUMMY	Defined
MBEDTLS_GENPRIME	Defined
MBEDTLS_NO_PLATFORM_ENTROPY	Defined
MBEDTLS_PK_RSA_ALT_SUPPORT	Defined
MBEDTLS_PKCS1_V15	Defined
MBEDTLS_PKCS1_V21	Defined
MBEDTLS_SELF_TEST	Defined
MBEDTLS_VERSION_FEATURES	Defined

Ciphering:

MBEDTLS_CIPHER_MODE_CBC	Defined
MBEDTLS_CIPHER_MODE_CFB	Defined
MBEDTLS_CIPHER_MODE_CTR	Defined
MBEDTLS_CIPHER_MODE_OFB	Defined
MBEDTLS_CIPHER_MODE_XTS	Defined
MBEDTLS_CIPHER_PADDING_PKCS7	Defined
MBEDTLS_CIPHER_PADDING_ONE_AND_ZEROS	Defined
MBEDTLS_CIPHER_PADDING_ZEROS_AND_LEN	Defined
MBEDTLS_CIPHER_PADDING_ZEROS	Defined
MBEDTLS_REMOVE_ARC4_CIPHERSUITES	Defined
MBEDTLS_REMOVE_3DES_CIPHERSUITES	Defined

Elliptic curves:

MBEDTLS_ECP_DP_SECP192R1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP224R1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP256R1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP384R1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP521R1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP192K1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP224K1_ENABLED	Defined
MBEDTLS_ECP_DP_SECP256K1_ENABLED	Defined
MBEDTLS_ECP_DP_BP256R1_ENABLED	Defined
MBEDTLS_ECP_DP_BP384R1_ENABLED	Defined
MBEDTLS_ECP_DP_BP512R1_ENABLED	Defined
MBEDTLS_ECP_DP_CURVE25519_ENABLED	Defined
MBEDTLS_ECP_DP_CURVE448_ENABLED	Defined

Key exchange:

MBEDTLS_KEY_EXCHANGE_PSK_ENABLED	Defined
MBEDTLS_KEY_EXCHANGE_DHE_PSK_ENABLED	Defined
MBEDTLS_KEY_EXCHANGE_ECDHE_PSK_ENABLED	Defined
MBEDTLS_KEY_EXCHANGE_RSA_ENABLED	Defined

MBEDTLS_KEY_EXCHANGE_DHE_RSA_ENABLED	Defined
MBEDTLS_KEY_EXCHANGE_ECDHE_RSA_ENABLED	Defined
MBEDTLS_KEY_EXCHANGE_ECDH_ECDSA_ENABLED	Defined
MBEDTLS_KEY_EXCHANGE_ECDH_RSA_ENABLED	Defined

SSL:

MBEDTLS_SSL_ALL_ALERT_MESSAGES	Not Defined
MBEDTLS_SSL_ENCRYPT_THEN_MAC	Not Defined
MBEDTLS_SSL_EXTENDED_MASTER_SECRET	Not Defined
MBEDTLS_SSL_FALLBACK_SCSV	Not Defined
MBEDTLS_SSL_RENEGOTIATION	Not Defined
MBEDTLS_SSL_PROTO_DTLS	Not Defined
MBEDTLS_SSL_DTLS_ANTI_REPLAY	Not Defined
MBEDTLS_SSL_DTLS_HELLO_VERIFY	Not Defined
MBEDTLS_SSL_DTLS_CLIENT_PORT_REUSE	Not Defined
MBEDTLS_SSL_DTLS_BADMAC_LIMIT	Not Defined

X509:

MBEDTLS_X509_CHECK_KEY_USAGE	Defined
MBEDTLS_X509_CHECK_EXTENDED_KEY_USAGE	Defined
MBEDTLS_X509_RSASSA_PSS_SUPPORT	Defined

3.29.3. Alternate implementation:

3.29.4. Modules:

General:

MBEDTLS_AESNI_C	Defined
MBEDTLS_AES_C	Defined
MBEDTLS_ARC4_C	Defined
MBEDTLS_ASN1_PARSE_C	Defined
MBEDTLS_ASN1_WRITE_C	Defined
MBEDTLS_BASE64_C	Defined
MBEDTLS_BIGNUM_C	Defined
MBEDTLS_BLOWFISH_C	Defined
MBEDTLS_CAMELLIA_C	Defined
MBEDTLS_CCM_C	Defined
MBEDTLS_CERTS_C	Defined
MBEDTLS_CIPHER_C	Defined
MBEDTLS_CHACHA20_C	Defined
MBEDTLS_CHACHAPOLY_C	Defined
MBEDTLS_CTR_DRBG_C	Defined

MBEDTLS_DES_C	Defined
MBEDTLS_DHM_C	Defined
MBEDTLS_ECDH_C	Defined
MBEDTLS_ECDSA_C	Defined
MBEDTLS_ECP_C	Defined
MBEDTLS_ENTROPY_C	Defined
MBEDTLS_ERROR_C	Defined
MBEDTLS_GCM_C	Defined
MBEDTLS_HKDF_C	Defined
MBEDTLS_HMAC_DRBG_C	Defined
MBEDTLS_MD_C	Defined
MBEDTLS_MD5_C	Defined
MBEDTLS_NIST_KW_C	Not Defined
MBEDTLS_OID_C	Defined
MBEDTLS_PADLOCK_C	Defined
MBEDTLS_PEM_PARSE_C	Defined
MBEDTLS_PEM_WRITE_C	Defined
MBEDTLS_PK_C	Defined
MBEDTLS_PK_PARSE_C	Defined
MBEDTLS_PK_WRITE_C	Defined
MBEDTLS_PKCS5_C	Defined
MBEDTLS_PKCS12_C	Defined
MBEDTLS_PLATFORM_C	Defined
MBEDTLS_RIPEMD160_C	Defined
MBEDTLS_POLY1305_C	Defined
MBEDTLS_RSA_C	Defined
MBEDTLS_SHA1_C	Defined
MBEDTLS_SHA256_C	Defined
MBEDTLS_SHA512_C	Defined
MBEDTLS_SSL_TICKET_C	Not Defined
MBEDTLS_SSL_TLS_C	Not Defined
MBEDTLS_VERSION_C	Defined
MBEDTLS_X509_USE_C	Defined
MBEDTLS_X509_CRT_PARSE_C	Defined
MBEDTLS_X509_CRL_PARSE_C	Defined
MBEDTLS_X509_CSR_PARSE_C	Defined
MBEDTLS_X509_CREATE_C	Defined
MBEDTLS_X509_CRT_WRITE_C	Defined
MBEDTLS_X509_CSR_WRITE_C	Defined
MBEDTLS_XTEA_C	Defined

3.29.5. Modules Configuration:

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC1	ADC1_INP11	Analog mode	No pull-up and no pull-down	n/a	
	PC5	ADC1_INP8	Analog mode	No pull-up and no pull-down	n/a	
	PB0	ADC1_INP9	Analog mode	No pull-up and no pull-down	n/a	
ADC2	PA7	ADC2_INP7	Analog mode	No pull-up and no pull-down	n/a	
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
FDCAN2	PB12	FDCAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB13	FDCAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
FMC	PI9	FMC_D30	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI10	FMC_D31	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF5	FMC_A5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC0	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH2	FMC_SDCKE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH3	FMC_SDNE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH8	FMC_D16	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH9	FMC_D17	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH10	FMC_D18	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH11	FMC_D19	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH12	FMC_D20	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH13	FMC_D21	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH14	FMC_D22	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH15	FMC_D23	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI0	FMC_D24	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI1	FMC_D25	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI2	FMC_D26	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI3	FMC_D27	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE0	FMC_NBL0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE1	FMC_NBL1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI4	FMC_NBL2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI5	FMC_NBL3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI6	FMC_D28	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI7	FMC_D29	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
I2C1	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB9	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
I2C2	PH4	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PH5	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
LTDC	PI11	LTDC_G6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI12	LTDC_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PI13	LTDC_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI14	LTDC_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF10	LTDC_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1	LTDC_R2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	LTDC_B5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB1	LTDC_R6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI15	LTDC_G2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ0	LTDC_R7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ2	LTDC_R3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ3	LTDC_R4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ4	LTDC_R5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ10	LTDC_G3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ11	LTDC_G4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PK0	LTDC_G5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PK2	LTDC_G7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA8	LTDC_B3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	LTDC_B4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ14	LTDC_B2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PK5	LTDC_B6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PK6	LTDC_B7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
PWR	PC2_C	PWR_CSTOP	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC3_C	PWR_CSLEEP	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA0	PWR_WKUP1	n/a	n/a	n/a	
	PA5	PWR_NDSTOP2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB7	PWR_PVD_IN	Analog mode	No pull-up and no pull-down	n/a	
QUADSPI	PE2	QUADSPI_BK1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF8	QUADSPI_BK1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB2	QUADSPI_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD12	QUADSPI_BK1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD13	QUADSPI_BK1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG6	QUADSPI_BK1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PC14-OSC32_IN (OSC32_IN)	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC1	PC8	SDMMC1_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC9	SDMMC1_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC10	SDMMC1_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC11	SDMMC1_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC12	SDMMC1_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD2	SDMMC1_CMD	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
SPI1	PA4	SPI1_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG11	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SPI6	PA15 (JTDI)	SPI6_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG14	SPI6_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB3 (JTDO/TRACESWO)	SPI6_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB4 (NJTRST)	SPI6_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM1	PK1	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PB14	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB15	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD6	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART3	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART6	PC6	USART6_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC7	USART6_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG12	USART6_RTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG13	USART6_CTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
Single Mapped Signals	PG2	FMC_A12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
GPIO	PE3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	
	PE4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	
	PC13	GPIO_EXTI13	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	
	PB5	GPIO_EXTI5	External Interrupt Mode with	No pull-up and no pull-down	n/a	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
			Rising edge trigger detection			

4.2. DMA configuration

nothing configured in DMA service

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
HASH and RNG global interrupts	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line3 interrupt	unused		
ADC1 and ADC2 global interrupts	unused		
FDCAN2 interrupt 0	unused		
FDCAN2 interrupt 1	unused		
EXTI line[9:5] interrupts	unused		
TIM1 break interrupt	unused		
TIM1 update interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
SPI1 global interrupt	unused		
USART1 global interrupt	unused		
USART2 global interrupt	unused		
USART3 global interrupt	unused		
EXTI line[15:10] interrupts	unused		
FMC global interrupt	unused		
SDMMC1 global interrupt	unused		
FDCAN calibration unit interrupt	unused		
USART6 global interrupt	unused		
CRYP global interrupt	unused		
FPU global interrupt	unused		
SPI6 global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
LTDC global interrupt	unused		
LTDC global error interrupt	unused		
QUADSPI global interrupt	unused		
HSEM1 global interrupt	unused		
Interrupt for all 6 wake-up pins	unused		

4.5.2. NVIC Code generation

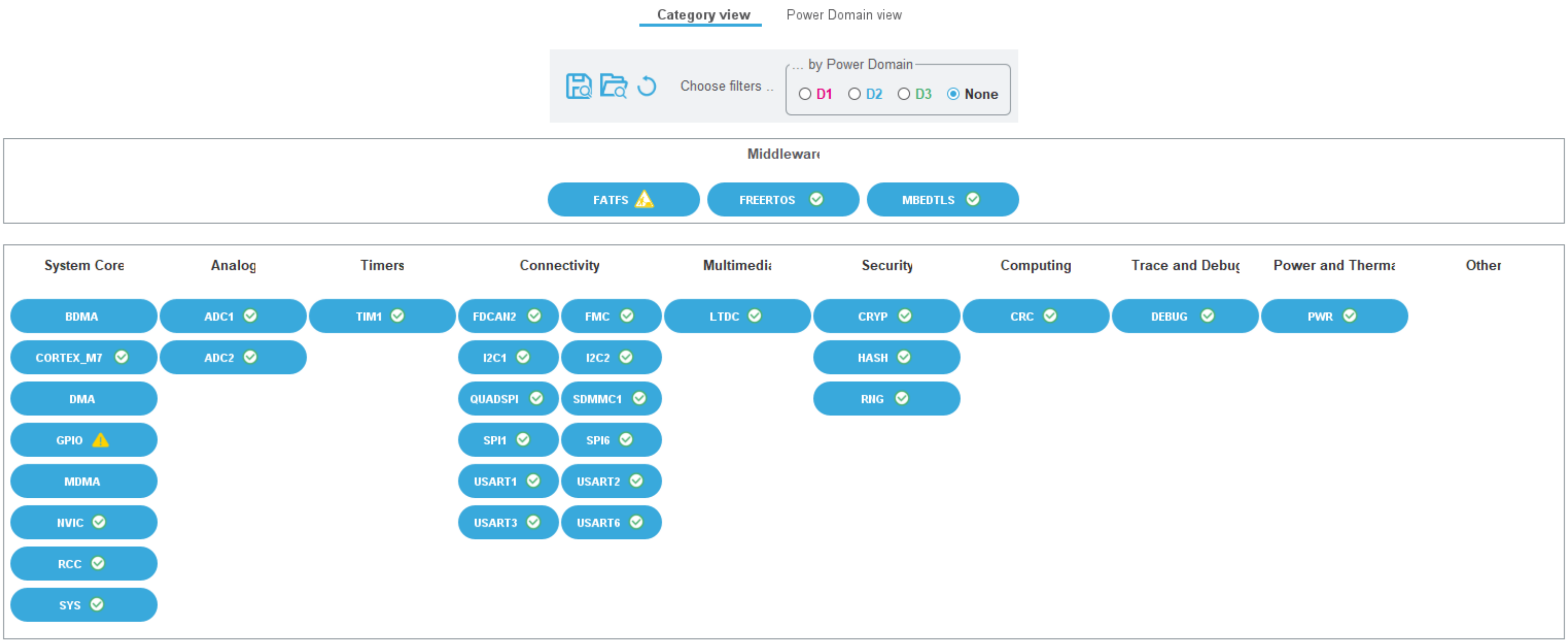
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	true	true
HASH and RNG global interrupts	false	true	true

*** User modified value**

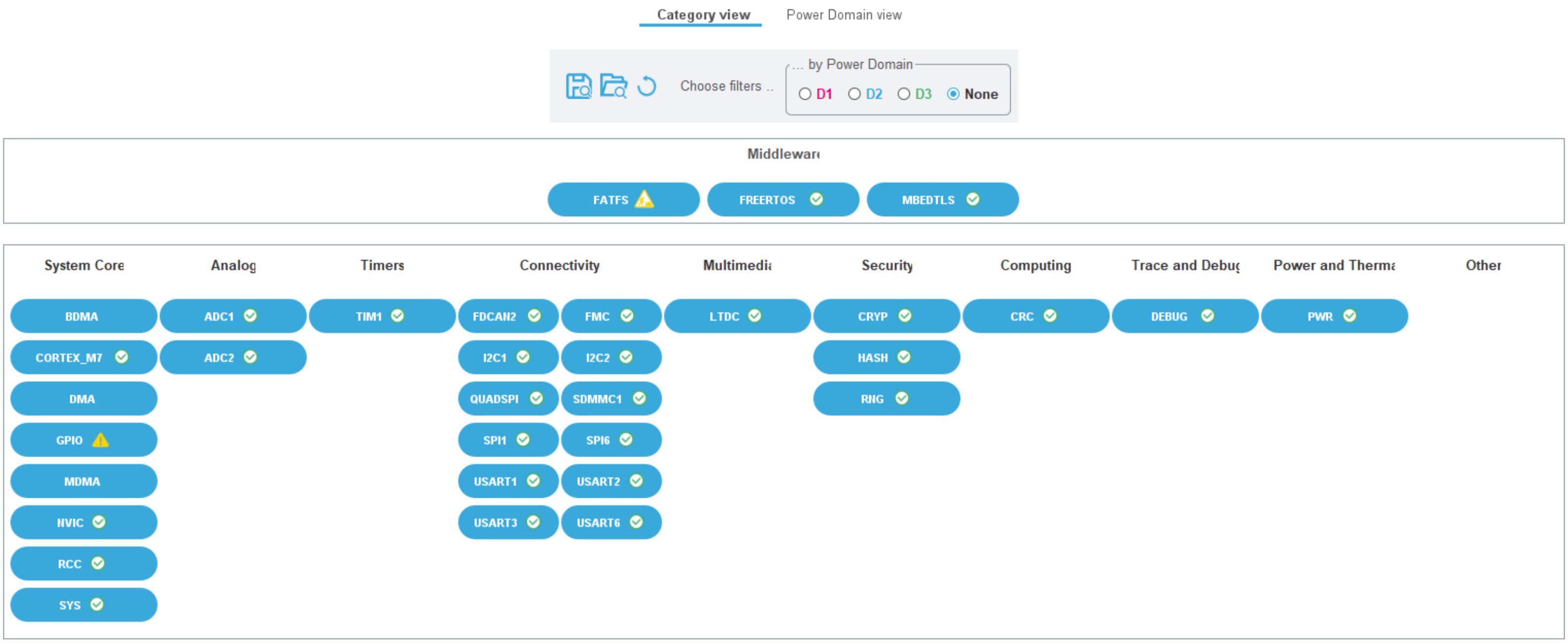
5. System Views

5.1. Category view

5.1.1. Current



5.1.2. Without filters



5.2. Power Domain view

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4936-migration-of-microcontroller-applications-from-stm32f7-series-to-stm32h743753-line-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5312-migration-from-revy-to-revv-for-stm32h743753-and-stm32h750-value-line-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors--stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4938-getting-started-with-stm32h74xig-and-stm32h75xig-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4992-introduction-to-secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4943-how-to-use-chromart-accelerator-to-refresh-an-lcdtft-display-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5507-how-to-use-crc-to-check-the-integrity-of-the-internal-flash-memory-on-stm32h7-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5337-guidelines-for-estimating-stm32h7-mcus-lifetime-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4996-how-to-use-the-jpeg-codec-peripheral-on-stm32-mcus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5200-getting-started-with-the-stm32h7-mcu-sdmmc-host-controller-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4323-getting-started-with-stemwin-library-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4435-guidelines-for-obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpd-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5450-stm32h7a37b3-lines-and-stm32h7b0-value-line-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
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